PHYSICAL AND CHEMICAL SCIENCE

Material Properties

Property words, sorting materials, magnetic materials and choosing the right material for the job



wood



key



glass bottle



shirt

In this pack

Learn it: Materials and their properties ★☆☆
Fact check: Sorting materials ★☆☆
Think it through: The right material for the job ★★☆
Investigate: Which material soaks up the most water? ★★☆
Real-world case study: Jute: the golden fibre ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

© InquiryMind · Learn · Play · Wonder · Free resources for everyone, bringing quality and equity to education and helping teachers build concepts and make learning visible · Illustrations drawn for InquiryMind · Pictures from OpenMoji (openmoji.org), CC BY-SA 4.0 · Print in black and white on A4 · Challenge: ★ core ★★ deeper ★★★ stretch

LEARN IT · C1 · MATERIALS

Materials and their properties

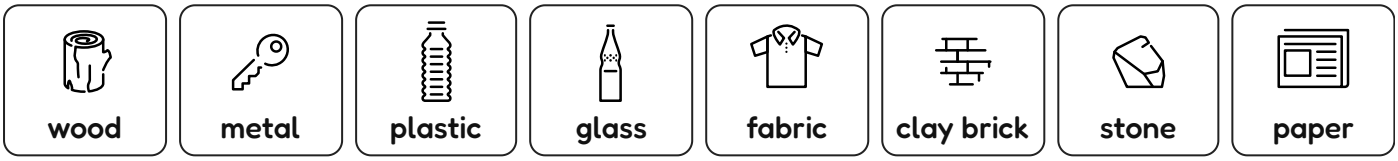
★☆☆ Grade 3

Name: Class: Date:

Learning goals

- Describe materials using property words: hard, soft, shiny, dull and more.
- Sort materials by their properties, including magnetic and non-magnetic.
- Explain why a material is chosen for a job.

Look around your classroom. The door is made of **wood**, the window of **glass**, the fan of **metal** and **plastic**, your shirt of **cotton**. These are all **materials**: what things are made of.



Properties

A **property** tells us what a material is like. Every material has its own properties.

Property words	Opposite	Example
hard: does not dent	**soft**: easy to squash	stone is hard; cotton wool is soft
shiny: reflects light	**dull**: not shiny	a steel spoon is shiny; wood is dull
flexible: bends easily	**stiff**: does not bend	rubber is flexible; a brick is stiff
transparent: you can see through it	**opaque**: you cannot see through it	glass is transparent; wood is opaque
waterproof: keeps water out	**absorbent**: soaks up water	plastic is waterproof; a towel is absorbent
strong: hard to break	**weak**: breaks easily	steel is strong; paper is weak

Magnetic or not?

A magnet pulls (attracts) things made of **iron** or **steel**. These are **magnetic** materials. Most materials are **not** magnetic: wood, plastic, glass, paper, fabric and even some metals such as **aluminium**, **copper**, gold and silver.

Try this at home

Use a fridge magnet to test things at home: a spoon, a key, a coin, a foil tray. Which ones stick?

Choosing the right material

We choose a material because its properties are right for the job.

Object	Material	Why?
window	glass	transparent, so light comes in
umbrella	plastic or waxed cloth	waterproof
cooking pot	metal (aluminium or steel)	strong and does not melt on the stove
pot handle	wood or plastic	does not get too hot to hold
towel	cotton	soft and absorbent
shopping bag	jute	strong, and it rots away when thrown out

Bangladesh connection

Bangladesh uses many natural materials: **jute** for bags and rope, **bamboo** for homes and bridges, **clay** for pots and bricks, and **cotton** for nakshi kantha quilts.

Key words

material what an object is made of
property what a material is like
transparent you can see through it
opaque you cannot see through it
waterproof does not let water through
magnetic pulled by a magnet

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C1 · MATERIALS

Sorting materials

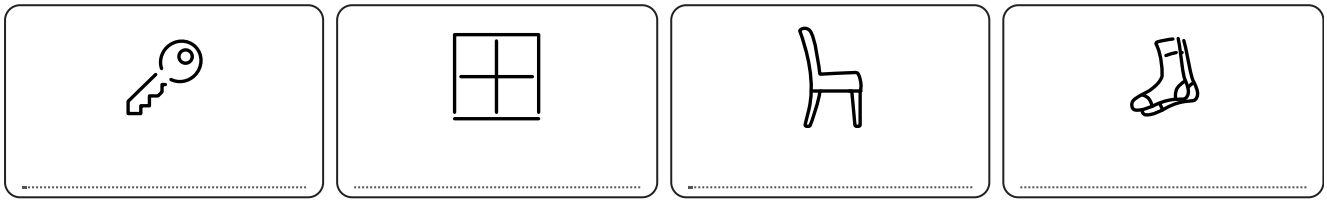
★☆☆
Grade 3

Name: Class: Date:

Learning goals

- Use property words.
- Sort materials.

1 Write the material each object is made of. [4]



2 Sort these materials. [6]

SORT THESE

steel stone cotton wool wool glass sponge

Hard	Soft

3 Will a magnet attract it? [5]

Object	Material	Magnetic?
nail	iron	
drink can	aluminium	
paper clip	steel	
ruler	plastic	
coin	brass or steel	

THINK IT THROUGH · C1 · MATERIALS

The right material for the job

★★☆
Grade 3

Name: Class: Date:

Learning goals

- Explain choices of materials.

1 **Why is a window made of glass, not wood?** [2]

2 **Why would a paper umbrella be a bad idea?** [2]

3 **Odd one out. Circle it and say why.** [2]

	A	B	C	D
group	iron nail	steel pin	aluminium can	paper clip

4 **Design a raincoat for a school trip. Label the materials and say why you chose them.** [3]

My raincoat

InquiryMind
LEARN · PLAY · WONDER

Which material soaks up the most water?

★★☆ Grade 3

Name: **Class:** **Date:**

Learning goals

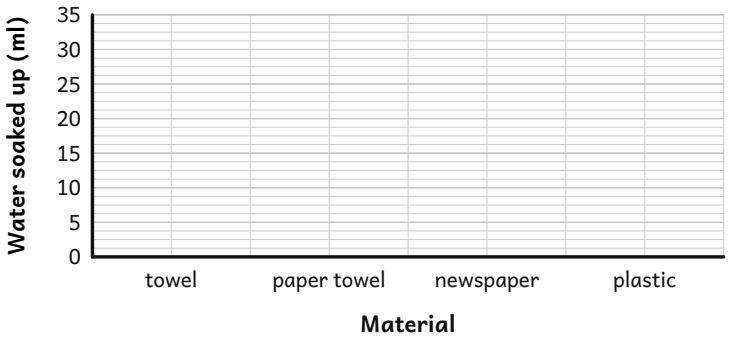
- Do a fair test.
- Draw a bar chart and conclude.

Class 3A put a 10 cm square of each material into a dish with 50 ml of water. After one minute they lifted it out and measured how much water was left in the dish.

1 **Complete the table: how much water did each material soak up?** [4]

Material	Water left (ml)	Water soaked up (ml)
cotton towel	20	
paper towel	30	
newspaper	40	
plastic	50	

2 **Draw a bar chart of the water soaked up.** [3]



3 **How did they make it a fair test?** [2]

.....

.....

4 **Which material would you use to mop up a spill? Why?** [1]

.....

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C1 · MATERIALS

Jute: the golden fibre

★★★
Grade 3

Name: **Class:** **Date:**

Learning goals

- See how properties make a material useful.

Science | Farming | Environment

Jute is a tall plant that grows well in the warm, wet fields of Bangladesh. Farmers soak the stems in water, then peel off long, shiny, golden threads called **fibres**. Because of its colour and value, jute is called the **golden fibre**.

Jute fibres are **strong** and **flexible**, so they are made into rope, sacks, mats and bags. Unlike plastic, jute **rots away** in the soil when it is thrown out. Bangladeshi scientists have also made a thin, plastic-like bag from jute, so that fewer plastic bags end up in rivers and drains.

Based on information about jute farming and jute products in Bangladesh.

1 **Write two properties of jute fibre.** [2]

2 **Why is a jute bag better for the environment than a plastic bag?** [2]

3 **Name two things made from jute.** [2]

Show what you know

★★★
Grade 3

Name: Class: Date:

Learning goals

- Use material words.

1 Find the words.

J	U	T	E	C	R	M	A	Y
S	H	O	O	Y	N	N	W	L
J	B	V	G	M	M	F	O	O
Y	S	E	W	L	E	I	O	I
R	N	H	O	R	A	T	D	N
E	G	H	I	S	U	S	A	D
U	T	N	A	N	O	I	S	L
U	T	C	J	R	Y	F	D	K
B	M	H	R	J	D	E	T	A

glass hard jute metal shiny soft wood

2 Which material is transparent? [1]

◯ wood ◯ glass

◯ brick

3 Which material is magnetic? [1]

◯ plastic ◯ aluminium

◯ iron

4 Challenge. A builder needs a material for a bridge. Which properties must it have? [2]

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
describe materials with property words	○	○	○
sort materials	○	○	○
test which materials are magnetic	○	○	○
explain why a material is chosen	○	○	○
run a fair test on materials	○	○	○